ICT2122 · Object Oriented Programming

Revision Guide and Practice

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

1. Main idea

Review encapsulation, abstraction, inheritance, polymorphism, interfaces, composition, and object identity. Prefer small public APIs, clear ownership, and tests of observable behavior over tests of private implementation details.

2. Example and reasoning

Practice: refactor a class with public mutable fields; decide whether inheritance or composition fits a notification service; add a boundary test; explain the Liskov substitution principle using a concrete subtype example.

3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

4. Questions for study

- Explain “revision guide and practice” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.